

phys. stat. sol. (a) **14**, 59 (1972)

Subject classification: 14.3.3; 13.3; 22.1.2

*Department of Electrical Engineering and Materials Research Laboratory,  
University of Illinois, Urbana***Current and Capacitance Transient Responses of MOS Capacitor<sup>1)</sup>****II. Recombination Centers in the Surface Space Charge Layer**

By

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The general theory of the high frequency capacitance transient for an initially depleted surface space charge layer with nonuniform distribution of bulk recombination centers is reduced to the forms suitable for determination of the spatial distribution of the recombination center concentration and the electric field dependences of the majority and minority carrier emission rates of the centers. Examples of gold diffused and control MOS capacitors (without gold diffusion) on n-type and p-type silicon are presented as illustrations.

Die allgemeine Theorie für das zeitliche Verhalten der Hochfrequenzkapazität für eine ursprünglich leere Oberflächenraumladungsschicht mit einer ungleichförmigen Verteilung der Volumenrekombinationszentren wird auf eine Form gebracht, die man für die Bestimmung der räumlichen Verteilung der Rekombinationszentren und der elektrischen Feldabhängigkeit der Emissionsraten der Minoritäts- und Majoritätsladungsträger aus den Zentren verwenden kann. Beispiele von MOS-Kapazitäten, in die Gold diffundiert wurde, und von MOS-Kapazitäten ohne Gold auf n- und p-leitendem Silizium werden zur Erläuterung angegeben.

**1. Introduction**

The transient characteristics of the current and high frequency capacitance of a MOS capacitor have been studied extensively<sup>2)</sup>. The general theory was developed and applied to the case of an initially depleted surface space charge layer [1]. The theory for the initially depleted surface is further developed in this paper and put into forms suitable for evaluating the properties of the recombination-generation-trapping centers in the surface space charge layer. Expressions for evaluating the spatial variation of the trap concentration,  $N_{TT}(x)$ , and the field dependences of the emission rates of the trapped majority carriers from phase I capacitance transient data and the field dependences of the emission rates of the trapped minority carriers from phase II capacitance transient data are obtained in Section 2. Experimental results for gold doped and non-gold doped control MOS capacitors on n-type and p-type silicon are presented in Sections 3 and 4.

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<sup>1)</sup> Supported in part by the Air Force Office of Scientific Research (Grant AFOSR-71-2067) and the Advanced Research Projects Agency.

<sup>2)</sup> For a historical review, see reference [1].

## 2. Theory

It was shown in [1] that the high-frequency capacitance transient of an initially depleted MOS capacitor can be divided into two phases, I and II, if the characteristic time constants of the two phases are substantially different. Phase I comes from redistribution of the trapped charges which induces a very small change in the space charge in the depleted surface space charge layer and a corresponding change in the layer width or the high-frequency capacitance. Phase I in a MOSC transient is entirely analogous and essentially identical to the entire transient in a p-n junction which is depleted of carriers in its space charge layer at all times. Phase II does not exist in a p-n junction but does exist in MOS capacitor due to the accumulation of the minority carriers at the insulator-semiconductor interface which are generated thermally, optically or by impact in the surface space charge layer. The accumulation of the minority carriers persists until the surface space charge layer reduces to the equilibrium or steady-state width. It is evident from these physical processes that phase I is mainly dependent on the emission of trapped majority carriers while phase II on the trapped minority carriers. This suggests that detailed evaluation of the phase I and II capacitance data should give the properties of both trapped majority and minority carriers.

### 2.1 Phase I

The phase I capacitance transient data can give both  $N_D(x)$  (we use the n-type silicon as the example) and  $N_{TT}(x)$  if the traps in the surface space charge layer is completely filled initially. This is well known for p-n junctions [2]. It can also give electric field dependence of the emission rate of the trapped majority carriers if the traps are completely filled in part of the surface space charge layer. The initial conditions, voltage waveforms and mathematics are given below for the MOS capacitor structure.

#### 2.1.1 Traps initially filled in the entire surface space charge layer

For this case, the applied voltage to the junction has the waveform shown in Fig. 1a and the corresponding high-frequency capacitance transient is shown in Fig. 1b. The flat-band voltage is assumed to be zero or alternatively the true applied gate voltage would be  $v_G = v_{G'} + V_{FB}$ . The traps are all filled by zero biasing or flat-banding the surface space charge region initially. After the gate voltage is stepped to a reverse bias value of  $V_2$ , the majority carriers will be

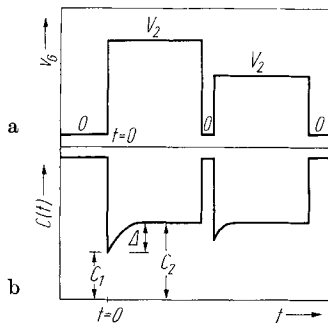


Fig. 1. a) Gate voltage and b) high-frequency capacitance waveforms during phase I when traps in the entire surface space charge layer are filled initially

swept out of the surface space charge region  $0 < x < w$  ( $t = 0$ )  $\equiv W_0$  in a transit time and the high-frequency capacitance is given by  $C_1 = C_o C_{s1}/(C_o + C_{s1})$ , where  $C_o = K_o \epsilon_0/x_0$  is the oxide capacitance and  $C_{s1} = K_s \epsilon_0/W_1$  is the surface space charge layer capacitance. The surface space charge layer width will decrease with time to a final value of  $w$  ( $\tau_I \ll t \ll \tau_{II}$ )  $= W_2$  due to the thermal emission of the trapped majority carriers in the surface space charge layer. This gives a final capacitance of  $C_2$  illustrated in Fig. 1b. The change of the capacitance,  $\Delta = C_2 - C_1$ , may be used to evaluate  $N_{TT}(W_1)$  as a function of  $W_1$  or  $W_2$  if data are taken over a range of  $V_2$ .

The expression for evaluating  $N_{TT}(W_1)$  from experimental data is obtained from the general result which relates the instantaneous applied voltage to the total space charge in the semiconductor<sup>3</sup>).

$$v_G' = - \int_0^\infty [(x_0/K_o \epsilon_0) + (x/K_s \epsilon_0)] \rho(x, t) dx, \quad (1)$$

where  $v_G' = v_G(t) - v_{FB}$ ,  $v_{FB} = \Phi_{MS} - (q_0 + q_{ss})/C_o$  and  $\rho(x, t) = q [p - n + N_D - n_T]$  for an n-type semiconductor with  $N_D(x)$  majority donors/cm<sup>3</sup> and  $n_T(x, t)$  trapped electrons/cm<sup>3</sup>. In the surface space charge layer of an initially depleted surface,  $0 < x < w(t)$ , we have  $p = n = 0$  for phase I and  $n_T(x, t) = N_{TT}(x) \exp(-t/\tau_t)$  if all the traps are filled initially. The phase I time constant is  $\tau_t = \tau_I = 1/(e_n + e_p) \approx 1/e_n$  for an n-type semiconductor since trap levels located above the intrinsic Fermi energy will have  $e_n \gg e_p$ . Here,  $N_{TT}(x)$  is assumed to be spatially varying such as due to the diffusion profile and both  $e_n$  and  $e_p$  are dependent on the local macroscopic electric field which lowers the potential well which the trapped electrons and holes have to surmount during thermal excitation. Thus, after the voltage step to a reverse bias of  $V_2$ , (1) becomes

$$V_2 = - \int_0^w [(x_0/K_o \epsilon_0) + (x/K_s \epsilon_0)] q [N_D(x) - n_T(x, t)] dx, \quad (2)$$

where we have neglected any time dependence of  $q_0$  and  $q_{ss}$ . The working expression for evaluating  $N_{TT}(x)$  and  $N_D(x)$  is obtained by varying  $V_2$ . Thus, taking the derivative of (2) with respect to  $V_2$ , and evaluating  $\partial w(t)/\partial V_2$  at  $t = 0$  and  $\tau_{II} \gg t \gg \tau_I$ , we have,

$$N_D(W_1) - N_{TT}(W_1) = (2/q K_s \epsilon_0)/(-\partial C_1^{-2}/\partial V_2) \quad (3)$$

and

$$N_D(W_2) = (2/q K_s \epsilon_0)/(-\partial C_2^{-2}/\partial V_2), \quad (4)$$

where  $W_1 = K_s \epsilon_0 (C_1^{-1} - C_o^{-1})$  and  $W_2 = K_s \epsilon_0 (C_2^{-1} - C_o^{-1})$ . Thus, from the experimental data of  $C_1$ ,  $C_2$ ,  $\partial C_1/\partial V_2$  and  $\partial C_2/\partial V_2$ , both  $N_D(W_1)$  and  $N_{TT}(W_1)$  can be obtained.

Since  $\Delta = C_2 - C_1$  is generally much smaller than  $C_1$  or  $C_2$  and it is easier to measure the small  $\Delta$  directly, the result given by (3) can be simplified to the following more convenient and exact form

$$N_{TT}(W_1) = -N_D(W_2) [1 - (\Delta/C_2)]^3/[1 - (\partial \Delta/\partial C_2)] + N_D(W_1). \quad (5)$$

Approximate solutions for special cases can be obtained. If  $N_D(W_2)$  is constant,

<sup>3</sup>) Equation (3.8) of reference [1].

then it would not be necessary to evaluate  $N_D(W_2)$  in (5) for each  $V_2$ . If  $N_D(x) \gg \gg N_{TT}(x)$ , then  $W_1 \approx W_2$  and  $\Delta \ll C_2$  so that (5) reduces to

$$N_{TT}(W_1)/N_D(W_2) \approx 3 (\Delta/C_2) - (\partial\Delta/\partial C_2) \approx 2 (\Delta/C_2) \approx N_{TT}(W_2)/N_D(W_2). \quad (6)$$

### 2.1.2 Traps initially filled only in part of the surface space charge layer

This initial condition will enable us to determine the electric field dependence of the thermal emission rate of the majority carriers during phase I. It is attained by an applied voltage wave form shown in Fig. 2a. The MOS capacitor is first flat-banded to deplete the minority carriers in the surface space charge region by zero bias,  $v_G = 0$ . Then a depletion voltage step,  $V_2$ , is applied for a duration until the entire phase I transient has passed so that the traps in the surface space charge layer,  $0 < x < W_2$ , are completely empty:  $n_T(x, t_2) = 0$ . At this time, the reverse bias is momentarily reduced to a value  $V_1$  (in the preceding case,  $V_1 = 0$ ) so that the traps in the region  $W_1 < x < W_2$  are now filled while traps in  $0 < x < W_1$  are still empty. The duration needs to be only long compared with the capture rate of the majority carriers which would be of the order of microseconds. Next, the bias is switched from  $V_1$  back to  $V_2$  and the capacitance transient is monitored.

The working equation can again be derived from (1) during the capacitance transient after  $t = 0$  labeled in Fig. 2b. The space charge density is given by  $\rho(x, t) = q N_D$  for  $0 < x < W_1$ ,  $\rho(x, t) = q (N_D - n_T)$  for  $W_1 < x < w(t)$ , and  $\rho(x, t) = 0$  for  $w(t) < x$ . This then gives an integral similar to that of (2).  $V_2$  is now kept constant while  $V_1$  is varied. Thus, taking the derivative  $(\partial/\partial V_1)$  and transforming  $w(t)$  to the measured high frequency capacitance  $C(t)$  by  $w(t) = K_s \epsilon_0 [C^{-1} - C_0^{-1}]$ , we have

$$n_T(W_1, t) = [n_T(w, t) - N_D(w)] [\partial C^{-2}/\partial C_1^{-2}], \quad (7)$$

where

$$n_T(x, t) = N_{TT}(x) \exp [-t/\tau_t(x)]. \quad (8)$$

The position dependence of  $\tau_t(x) = 1/(e_n + e_p)$  comes from the electric field dependence of  $e_n = e_n(E)$  and  $e_p = e_p(E)$  and the electric field varies with position  $E = E(x)$  in the depleted surface space charge layer. We note immediately that at  $t = 0$ , (7) will give another solution for evaluating  $N_{TT}(x)$  and  $N_D(x)$  which is

$$N_{TT}(W_1) = [N_{TT}(W_1) - N_D(W_1)] [\partial C_1^{-2}/\partial C_1^{-2}]. \quad (9)$$

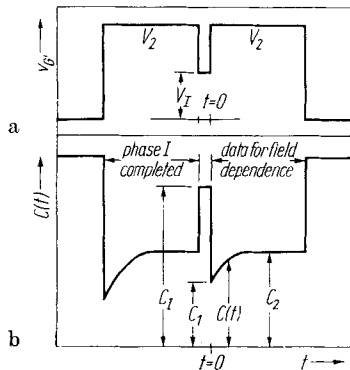


Fig. 2. a) Gate voltage and b) high-frequency capacitance waveforms during phase I when traps in part of the surface space charge layer are filled initially

This is a more difficult measurement than (5) since the changes in  $C_1$  and  $C_I$  for a change in  $V_I$  are both small while  $C_1$  and  $C_I$  may be widely different. In (5) both the changes in  $C_1$  and  $C_2 = C_1 + \Delta$  or  $\Delta$  need to be measured, however,  $C_1$  and  $C_2$  are nearly equal, making  $C_1$ ,  $C_2$  and  $\Delta$  measurements more accurate.

To obtain the field dependence of the time constant  $\tau_t$  or the thermal emission rates, (7) can be differentiated with respect to time and the results evaluated at some predetermined time  $t$ . A simple solution is obtained if  $N_{TT} \ll N_D$  and  $N_D(w)$  does not vary appreciably with position so that  $\partial N_D(w)/\partial t = [dN_D(w)/dw] (\partial w/\partial t)$  can be neglected. For this case, we have

$$[e_n(W_I) + e_p(W_I)] N_{TT}(W_I) = N_D(w) (\partial/\partial C_I^{-2}) (\partial C^{-2}/\partial t). \quad (10)$$

Alternatively, we may use a graphical procedure by subtracting two capacitance transients for two slightly different  $V_I$ :  $V_{I1}$  and  $V_{I2} = V_{I1} + \Delta V_I$  at constant times. From (7), we get immediately

$$\begin{aligned} n_T(W_I, t) &= N_{TT}(W_I) \exp [-t/\tau_t(W_I)] \\ &= [n_T(w, t) - N_D(w)] (C_I/C)^3 (\Delta C/\Delta C_I) \end{aligned} \quad (11)$$

$$\approx -N_D(W_2) (C_I/C_2)^3 \Delta C/\Delta C_I \quad N_{TT} \ll N_D. \quad (12)$$

Thus, the difference of the two capacitance transient decay curves  $\Delta C(t)$ , would be truly exponential as indicated in (12) if  $N_{TT} \ll N_D$ . The time constant would be the value evaluated at  $W_I$  or at the electric field at  $W_I$ . The emission rates at the electric field  $E_I$  ( $x = W_I$ ) for an applied reverse bias of  $V_2$  can then be obtained from  $e_n(E_I) + e_p(E_I) = 1/\tau_t$ . It is evident in (10) to (12) that in order to get high resolution of the electric field dependence,  $V_I$  or  $C_I$  should be made small. However, this would produce larger error in subtracting two decay curves to get  $\Delta C(t)$ . In addition, if  $N_{TT} \ll N_D$  is not satisfied, (11) shows that an important correction must be made when (12) is used. These methods have been applied to silicon p-n junctions to obtain the electric field dependences of the thermal emission rates at sulfur [3], zinc [4], silver and cobalt [5] centers.

## 2.2 Phase II

The general solution for p-type silicon during phase II is<sup>4)</sup>

$$\begin{aligned} C_o \dot{v}_G + \dot{q}_O - i_{PS} - j_{NW} + \int_0^w i_N dx - \\ - (\partial/\partial t) \int_0^w q (N_A + n_T) [1 + (K_o x/K_s x_o)] dx = 0, \end{aligned} \quad (13)$$

where ion migration in the oxide,  $\dot{q}_O$ , trapping at the interface states,  $i_{PS}$ , and the generation current in the quasi-neutral region,  $j_{NW}$ , are all neglected in this work. Since  $\dot{n}_T = 0$  for phase II and  $n = p = 0$  in  $0 < x < w$ , the volume generation current is given by its steady-state value:  $i_N(x, t) = -q e_n n_T = -q N_{TT} e_n e_p/(e_n + e_p)$ . These simplify (13) to

$$\int_0^w [e_n e_p/(e_n + e_p)] N_{TT}(x) dx + (\partial/\partial t) \int_0^w (N_A + N_T) [1 + (K_o x/K_s x_o)] dx = 0, \quad (14)$$

<sup>4)</sup> Equation (3.17) of reference [1].

where  $N_T = N_{TT} e_p / (e_n + e_p)$  is the steady-state trapped electron concentration. It is immediately evident that trap properties can be evaluated from the first integral if a partial derivative,  $\partial(14)/\partial w$ , is made. Using  $w(t) = K_s \epsilon_0 (C^{-1} - C_0^{-1})$ , where  $C(t)$  is the experimental data, then writing  $\xi \equiv C_0/C$ , we have

$$[e_n e_p / (e_n + e_p)]_w N_{TT}(w) = \frac{1}{2} [N_A(w) + N_T(w)] [ - (\partial/\partial \xi) (\partial \xi^2 / \partial t) ] + (K_s x_0 / K_0) \{ (\partial/\partial w) \ln [N_A(w) + N_T(w)] \} ( - \partial \xi^2 / \partial t) . \quad (15)$$

Since  $e_p \gg e_n$  for trap levels in the lower half gap in p-type silicon, and if  $N_{TT}(w) \ll \ll N_A(w)$ , then (15) can be simplified to

$$2 e_n(w) N_{TT}(w) / N_A(w) \approx - (\partial/\partial \xi) (\partial \xi^2 / \partial t) - (K_s x_0 / K_0) (\partial \ln N_A / \partial w) (\partial \xi^2 / \partial t) \quad (16)$$

$$\approx - (\partial/\partial \xi) (\partial \xi^2 / \partial t) , \quad (17)$$

where (17) applies when the substrate dopant impurity concentration does not vary substantially in the surface space charge layer. For n-type semiconductor, (16) and (17) become

$$2 e_p(w) N_{TT}(w) / N_D(w) \approx - (\partial/\partial \xi) (\partial \xi^2 / \partial t) - (K_s x_0 / K_0) (\partial \ln N_D / \partial w) (\partial \xi^2 / \partial t) \quad (18)$$

$$\approx - (\partial/\partial \xi) (\partial \xi^2 / \partial t) . \quad (19)$$

These results show that the minority carrier emission rate is that at the edge of the depletion layer,  $x = w$ , where the electric field is low and can be estimated by the value where  $V(x) = \Phi_F$ . Thus,  $e_p(w)$  or  $e_n(w)$  in (16) to (19) are essentially constant and nearly equal to the low field or equilibrium value, if  $N_D$  or  $N_A$  is low.

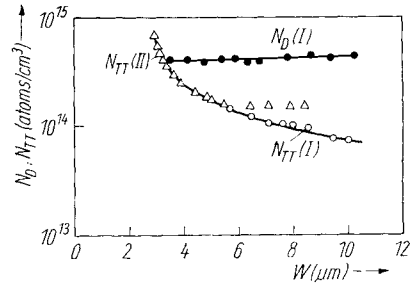
### 3. Experiments on n-type Silicon

Experimental data are analysed for both gold diffused and control (no gold diffusion) MOS capacitors on both n- and p-type silicon substrates or silicon epitaxial layers. Both diffused majority carrier and MOS gate types of guard rings around the 30 mil diameter MOS capacitor are used to cut off surface channel which is a source of minority carrier. The presence of channels would hinder the interpretation of the phase I and II transients. This section will discuss results on n-type silicon which represents a typical single level situation located near midgap. The p-type silicon diffused with gold is discussed in the next section which is a typical two level case with the shallow level gold donor active during phase I and the midgap gold acceptor level active during phase II. Surface channel effects on the capacitance transients will be discussed elsewhere.

#### 3.1 Gold diffused capacitor (GI-B)

Non epitaxial n-type, float zone (111) silicon of  $N_D = 4 \times 10^{14}$  phosphorus/cm<sup>3</sup> from Wacker Chemical of about 15 mil thickness is chemically polished to a mirror finish and then oxidized in a dry oxygen ambient at 1000 °C to give an oxide thickness of approximately 2000 Å. High purity gold (6-9) is then evaporated onto one of the surfaces where the oxide was removed and diffused into the slice at 850 °C for 40 min in a dry argon ambient. Aluminum gate of 30 mil diameter with 2 mil wide MOS guard ring of 32 mil inner diameter are photoengraved onto the oxidized side of the surface. Gold on the back is then lapped off and square dices each containing four devices are scribed and broken

Fig. 3. Impurity profile for gold diffused n-Si MOS capacitor. Phase I data are  $N_D(I)$ ,  $N_{TT}(I)$  and phase II data are  $2e_p^t N_{TT}/N_D$  and computed  $N_{TT}(II)$  from  $N_D(I)$  and  $e_p^t = 3.4 \times 10^{-3}/s$  at 215.5 °K

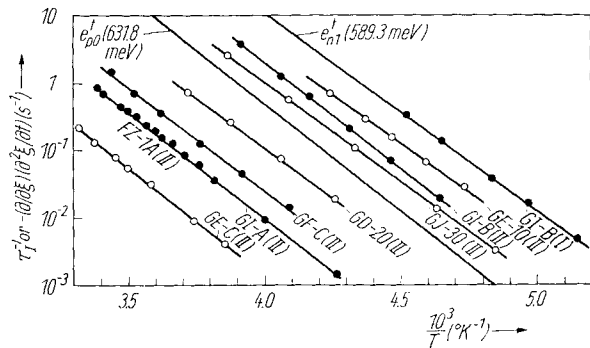


off from the slice. These are then mounted in TO-5 8-pin headers and ultrasonically lead bonded. A can is pressure fitted to the header. The capacitance transient measurements were performed at 1 MHz using a Boonton capacitance meter with an analog output whose response time is approximately 1 ms. The canned header is mounted on a copper block with a dc heater surrounding the block. The tail end of the copper block is immersed in a liquid nitrogen dewar. Temperature of the sample is automatically controlled by comparing the thermocouple output attached to the copper block with a reference voltage. The error is amplified by a HP-425 microvoltmeter whose output drives a dc power supply which powers the heater surrounding the copper block. Temperature stability can be maintained to within 5  $\mu$ V or better using copper-constantan thermocouple which corresponds to about  $\pm 0.2$  °K or better above 77 °K. Using the liquid nitrogen in the dewar for the reference junction and a calibrated 5-digit digital voltmeter or a L&N potentiometer, absolute temperature reading can be obtained to about the same accuracy as the stability.

The phase I capacitance transient of a typical MOS capacitor in this group is taken at 215.5 °K where  $\tau_I$  is about 7.7 s. Gate voltage is switched from an initial value of +10 V to achieve flat band or accumulation to a final value  $V_2$  from +5 to -70 V to cover a range of surface space charge layer width of 1 to 10  $\mu$ m. The concentration profile of the donor,  $N_D(W)$ , using (4) obtained from the data is shown in Fig. 3. The total capacitance change,  $\Delta$ , is also measured for each  $V_2$  and the gold concentration profile is obtained from (5) and shown also in Fig. 3 as circles from 5 to 10  $\mu$ m.

The phase I capacitance transients are also obtained for four other temperatures, 194.6, 201.3, 207.0, and 221.0 °K. The semilog plots of  $\Delta C$  vs.  $t$  are all straight lines and the phase I time constants thus obtained give the trapped

Fig. 4. Reciprocal phase I and II time constants,  $e_n^t$  or  $e_p^t$  for phase I and  $2e_p^t N_{TT}/N_D$  or  $2e_n^t N_{TT}/N_A$  for phase II for all devices presented in this paper. The two solid lines are  $e_{n1}^t$  and  $e_{p0}^t$  at the gold acceptor level from reference [6]. ● p-Si, ○ n-Si,



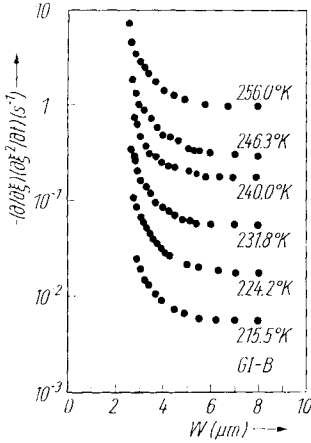


Fig. 5.  $-(\partial/\partial\xi)(\partial\xi^2/\partial t)$  or approximately  $2 e_p^t \times \times N_T/N_D$  for the gold diffused n-Si MOS capacitor unit GI-B as a function of position at six temperatures

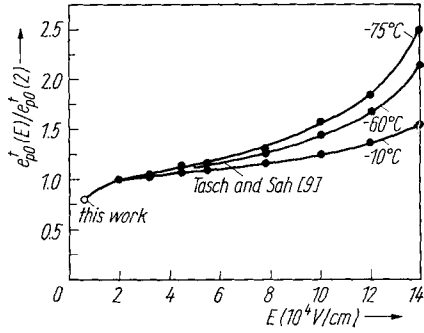


Fig. 6. The electric field dependences of  $e_{p0}^t$  at the gold acceptor level in silicon from Tasch and Sah [9] and one point with the lowest field obtained from phase II data

electron thermal emission rate at the gold acceptor center:  $\tau_1 = 1/e_{n1}^t$ . The emission rates are plotted as a function of  $1000/T$  °K in Fig. 4 where the two solid lines are those determined previously [6]. It is seen that excellent agreement between these MOSC data and previous p-n junction data is obtained.

The phase II data shown in Fig. 3 and 4 are obtained as follows. The gate bias voltage is switched from +10 to -20 V and the capacitance transients are obtained on a strip chart recorder at six temperatures which are above the range of the phase I data. The capacitance data are analysed using (19) since the phase I data showed that  $N_D$  is essentially constant. The results are shown in Fig. 5 which show that all the curves are parallel as we would expect. Since  $e_p^t(W)$  is the value at the edge of the depleted surface charge layer where the field is low, it is relatively independent of  $W$ . Thus, the variation with  $W$  indicated in Fig. 5 reflects entirely that of  $N_{TT}(W)$  for the constant  $N_D$  in this case. Based on this, the 215.5 °K phase II curve of Fig. 5 is matched to the  $N_{TT}(W)$  data of phase I in Fig. 3, giving a value of  $e_p^t(W) = 3.4 \times 10^{-3}/s$  compared with the p-n junction data of  $4.3 \times 10^{-3}/s$  previously obtained [6]. The smaller value here reflects the field dependence of  $e_p^t$ . The present value from MOSC is approximately at an electric field of  $6 \times 10^3$  V/cm while the lowest field in the previous work was about 1 to  $2 \times 10^4$  V/cm. A plot of the present and previous data of field dependences is given in Fig. 6, which shows that the equilibrium value is probably not reached even at  $6 \times 10^3$  V/cm in the present case.

The higher apparent value of  $N_{TT}$  from phase II in  $5 < W < 10 \mu m$  in Fig. 3 compared with those from phase I is not due to field dependence since  $N_{TT}(W, \text{phase II})$  corresponds to the edge or low field value. Instead, the higher apparent  $N_{TT}(W, \text{II})$  at large  $W$  comes from the interband impact ionization (secondary multiplication) in the high-field surface space charge region around the perimeter of the MOS capacitor.

The thermal activation energy of  $e_p^t(W)$  from the phase II data in Fig. 5 is obtained in the  $1000/T$  plot in Fig. 4 at  $W = 3 \mu m$ . It is 630 meV which is identical to that obtained from p-n junctions [6]  $e_{p0}^t = (2.58 \pm 0.27) \times 10^{12} \exp [(-631.8 \pm 2.2)/(kT)]$ .



These results not only verify previous junction data [6] in contrast to some recent work from junction conductance measurements [7] but also give a detailed gold diffusion profile in silicon oxide surface layer.

### 3.2 Control capacitors (GI-A, GF-C, FZ-1A)

Three types of control capacitors without gold diffusion are studied. GI-A is similar to GI-B just discussed but without gold diffusion. GF-C has diffused N+ guard ring instead of MOS guard ring and is made on epitaxial film of  $N_D = 5.6 \times 10^{15}$  phosphorus/cm<sup>3</sup>. FZ-1A has no guard ring and uses crucible grown silicon of  $N_D = 5.1 \times 10^{14}$  phosphorus/cm<sup>3</sup>.

The phase I capacitance transients were too small and not measured. The phase II capacitance transients are taken over a range of temperature for each device such that the time required for the entire curve lies between 10 s and 1 h. These data are then analyzed according to (19) and plotted as a function of  $1000/T$  °K in Fig. 4. Note that the epitaxial sample (GF-C) with phosphorus diffused N+ guard ring has about a factor of two higher  $e_p N_{TT}/N_D$  than the other two nonepitaxial and none phosphorus diffused samples (GI-A and FZ-1A). This is mainly from process variability and illustrates that the phase II transient can be used to monitor and study in detail the process parameters.

Another interesting feature which is useful in application is shown in Fig. 7 for the unit FZ-1A at 271.0 °K for three depletion voltages, -10, -20, and -30 V. This is the spike which appears at  $W = 2.5 \mu\text{m}$  and implies a high value of  $e_p N_{TT}/N_D$  at this point in the sample, or the presence of a localized defect with high density of traps,  $N_{TT}$ , or high carrier generation rate,  $e_p$ , or low value of  $N_D$ . A similar hump appears in the phase II data of the float zone device, GI-A, at about  $2.8 \mu\text{m}$ .

A third feature of the phase II data in Fig. 7 is the rapid rise of  $e_p N_{TT}/N_D$  in the interior. Since  $N_D$  is essentially constant from phase I data such as that shown in Fig. 3 and  $N_{TT}$  decreases going away from the surface  $W = 0$ , this rise must indicate an increase of  $e_p$  with  $W$  or with increasing electric field. Since  $e_p = e_p(W)$  is the low-field value as shown in Section 2, the apparent increase must be due to the neglected interband impact generation of electron-hole pairs (secondary multiplication) in the high-field region of the surface space charge layer, such as the perimeter. It may also be due to the 'softness' of the surface space charge layer from metallic precipitates, although in this case,  $e_p N_{TT}/N_D$  vs.  $W$  curves should be more random than observed.

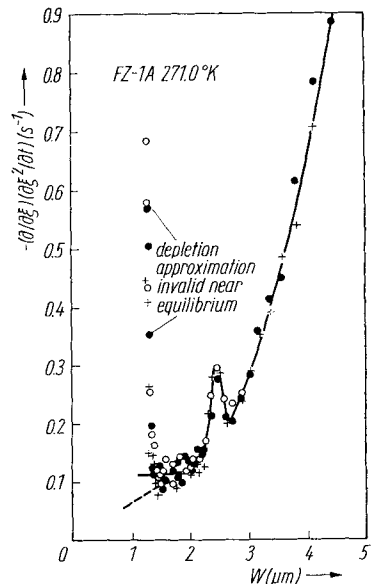


Fig. 7. Phase II data of an n-Si control MOS capacitor showing a hump from localized defect.  $-(\partial/\partial\xi)(\partial\xi^2/\partial t) \approx 2 e_p \times N_{TT}/N_D$ ,  $\circ$  -10 V,  $\bullet$  -20 V,  $+$  -30 V

## 4. Experiments on p-Type Silicon

### 4.1 Gold diffused capacitors (GD-20, GE-10, GJ-30)

GD-20 and GE-10 have epitaxial film of  $N_A = 2.5 \times 10^{15}$  boron/cm<sup>3</sup>. GD-20 is gold diffused at 1000 °C for 20 min and has no guard ring while GE-10 for 10 min and has also a P+ boron diffused guard ring. GJ-30 has crucible grown (111) silicon substrate with  $8.6 \times 10^{14}$  boron/cm<sup>3</sup>. It is gold diffused at 900 °C for 30 min, after which the oxide on the top surface is etched off and regrown at 875 °C for 2 h in oxygen bubbled through 95 °C water. This is designed to illustrate outdiffusion of gold during oxidation which will be revealed in the phase I and II data.

Detailed phase I data are taken for GJ-30 at 121.2 °K which gives a convenient  $\tau_i$  of about 20 s due to hole emission from the  $E_V + 380$  mV gold donor level. The data are analysed according to (4) and (5) and plotted in Fig. 8. Considerable field dependences are observed and analysed according to (10) and (11), giving a 20% increase of  $e_{p-1}^t$  at the gold donor level when the electric field is increased from  $8 \times 10^3$  to  $3.5 \times 10^4$  V/cm. The temperature dependence of the phase I time constant,  $\tau_I = 1/e_{p-1}^t$ , is also taken at the lowest field and shown in Fig. 9, in excellent agreement with previous results from p-n junctions [6], which is shown as solid line.

The phase II data for GJ-30 at 230.5 °K ( $e_{n1}^t = 1.16$ /s for the gold acceptor level [6]) and 259.1 °K ( $e_{n1}^t = 30.4$ /s) [6] are analysed according to (19) and shown in Fig. 8 using the quoted values from reference [6] just given for  $e_{n1}^t$  in  $2e_{n1}^t N_{TT}/N_A$  to get  $N_{TT}/N_A$  from the experimental  $-(\partial/\partial\xi)(\partial\xi^2/\partial t)$ . Since  $N_A$  is essentially constant from the phase I data shown in Fig. 8, the approximation of (19) is used. An obviously very interesting feature is exhibited in Fig. 8: the gold concentration profile has a minimum at about 1.2  $\mu$ m from the SiO<sub>2</sub>-Si interface. This comes about due to etching off the original oxide and reoxidation at a low temperature (875 °C) subsequent to the 900 °C gold diffusion.

The phase II data for GD-20 and GE-10 were also obtained, showing essentially constant  $2e_n^t N_{TT}/N_A$  beyond 1  $\mu$ m from the surface and rapid increase below 1  $\mu$ m which comes partly from depletion of  $N_A$  near the surface during

thermal oxidation and pile-up of gold at the interface, with the latter similar to those observed for n-Si in Fig. 3 and 5. A profile plot is given in Fig. 10.

Temperature dependences of the phase II capacitance transients are obtained for all three types of samples. The data,  $-(\partial/\partial\xi)(\partial\xi^2/\partial t) \approx 2e_n^t N_{TT}/N_A$ , in the interior where it is independent of position are plotted in Fig. 9 and labeled GE-10, GJ-30, and

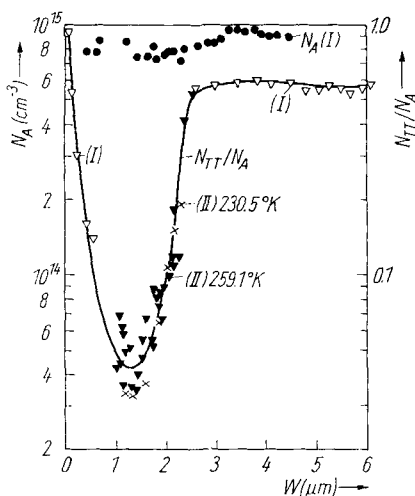


Fig. 8. Impurity profiles for a gold diffused p-Si MOS capacitor whose oxide is regrown at 875 °C in wet (95 °C water) O<sub>2</sub> after gold diffusion and original oxide is etched off. Phase I data:  $N_A(I)$  and  $(N_{TT}/N_A)(I)$  at 121.1 °K. Phase II data:  $(N_{TT}/N_A)(II)$  at two temperatures calculated from  $2e_n^t N_{TT}/N_A$  and using  $e_n^t(259.1 \text{ °K}) = 30.4$ /s and  $e_n^t(230.5 \text{ °K}) = 1.16$ /s from reference [6].

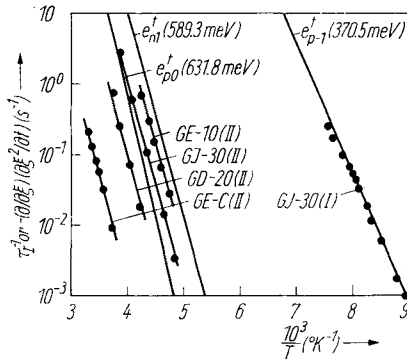


Fig. 9. Reciprocal time constants of phase I and II transients,  $e_{p-1}^t$  and  $2e_n^t N_{TT}/N_A$ , for both gold diffused (GJ-30, GE-10, GD-20) and control (GE-C) MOSC on p-Si.

$$-(\partial/\partial\xi) (\partial\xi^2/\partial t) \approx 2(W) e_n^t N_{TT}(W)/N_A(W)$$

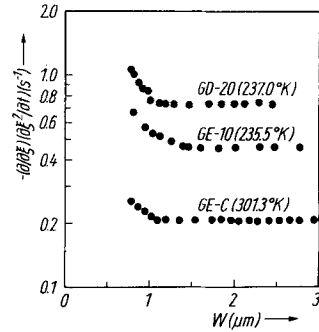


Fig. 10. Phase II data of three gold diffused MOSC on p-Si.  $-(\partial/\partial\xi) (\partial\xi^2/\partial t) \approx 2e_n^t(W) N_{TT}(W)/N_A(W)$

GD-20. They all have the thermal activation energy of the electron emission rate at the gold acceptor level, 589 meV. This indicates that phase II is controlled by the gold acceptor level near the midgap in p-Si while phase I was shown to be due to the hole emission from the gold donor level at  $E_V + 380$  meV. This is expected since the phase II transient from thermal emission of minority carriers (electrons) trapped at the gold donor level in p-Si cannot be observed due to too small a thermal emission rate from the large thermal activation energy,  $E_C - E_T = E_G - 370 = 800$  meV compared with the majority carrier (hole) and minority carrier (electron) emission rates at the midgap acceptor level.

The relative magnitude of  $e_n^t N_{TT}/N_A$  for GD-20 and GE-10 indicated in Fig. 9 (GE-10 is 50 times higher than GD-20) appears to indicate the effect of p+ boron diffusion from a BN source to form the guard ring in the GE-10 sample, however, more detailed experiments using this technique are needed. These phase II data are also plotted in Fig. 4 for comparison with those from n-Si obtained in Section 3.

#### 4.2 Control capacitor (GE-C)

This device is identical to the GE-10 device just discussed except that no gold diffusion is made. The phase I capacitance transients are again too small to be measured. Phase II transients are obtained over a temperature range just below room temperature and a typical plot is shown in Fig. 10, labeled GE-C (301.3 °K). Since there is no intentionally diffused recombination impurity, the rise of  $2e_n^t N_{TT}/N_A$  towards the interface,  $W = 0$ , indicated in Fig. 10 is probably due to out diffusion of boron during thermal oxidation [8] with the consequence of decreasing  $N_A$  towards the interface.

The temperature dependences of the phase II transient are shown both in Fig. 9 and 4 by the lines labeled GE-C. Fig. 4 shows that the p-Si control sample (GE-C) has nearly the same  $e_n^t N_{TT}$  as the  $e_p^t N_{TT}$  of the two lowest n-Si control samples (GI-A and FZ-1A) if the ratio  $N_A/N_D = 2.5 \times 10^{15}/4 \times 10^{14}$  is taken into account. This suggests that the residue recombination-generation level for phase II is probably an intrinsic bulk or lattice defect located at midgap whose thermal activation energy is approximately 600 meV.

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*(Received June 19, 1972)*